Setwise Smoothness of an RKHS Automatically Gives a Continuous Embedding

Automatic math-discovery run `fa_banach_001`

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Source Problem

The source is Carl-Johann Simon-Gabriel and Bernhard Schölkopf, *Kernel Distribution Embeddings: Universal Kernels, Characteristic Kernels and Kernel Metrics on Distributions*, arXiv:1604.05251 [1]. After Proposition 4, the authors ask whether, on page 9, mere set inclusion of the RKHS in a smooth function space forces continuous inclusion. Their exact question is:

> "We do not know whether, in general, $\mathcal{H}_k \subset C^m_*$ implies $\mathcal{H}_k \hookrightarrow C^m_*$."

Proposition 4 (Sufficient condition for $\mathcal{H}_k \hookrightarrow \mathscr{C}^m_\star$).

If $k \in \mathscr{C}^{(m,m)}$ *, then* $\mathcal{H}_k \hookrightarrow \mathscr{C}^m$.
If $k \in \mathscr{C}^{(m,m)}_0$*, then* $\mathcal{H}_k \hookrightarrow \mathscr{C}^m_0$.
If $k \in \mathscr{C}^{(m,m)}_b$*, then* $\mathcal{H}_k \hookrightarrow \mathscr{C}^m_b$*, thus* $\mathcal{H}_k \hookrightarrow (\mathscr{C}^m_b)_c$.

Proposition 3 shows that $\mathcal{H}_k \subset \mathscr{C}_*$ iff $\mathcal{H}_k \hookrightarrow \mathscr{C}_*$. As there exist non-continuous kernels whose RKHS functions are all continuous, the hypothesises in Proposition 4 is sufficient but not necessary. We do not know whether, in general, $\mathcal{H}_k \subset \mathscr{C}^m_*$ implies $\mathcal{H}_k \hookrightarrow \mathscr{C}^m_*$. However, if $\mathcal{H}_k \subset \mathscr{C}^{m+1}$, then $k \in \mathscr{C}^{(m,m)}$ (Grothendieck, 1953, cited by Schwartz, 1954, Lemme 2), thus $\mathcal{H}_k \hookrightarrow \mathscr{C}^m$. In

In their notation, the symbol C^m_* means either $C^m(\Omega)$, with uniform convergence on compact sets of all derivatives through order m, or $C^m_0(\Omega)$, with uniform convergence on all of Ω and all those derivatives vanishing at infinity. Here $\Omega \subset \mathbb{R}^d$ is open, and $m \in \mathbb{N}_0 \cup \{\infty\}$.

Result Type

This packet gives a **full affirmative answer**. No smoothness or even continuity assumption on the kernel is required beyond the stated set inclusion.

Proof Intuition

The only possible obstruction is discontinuity of the set-theoretic identity map from the RKHS norm to the smooth topology. But such discontinuity cannot hide in a sequence with two different limits. RKHS-norm convergence always determines the pointwise limit, because point evaluation is bounded; smooth target convergence also determines the pointwise limit, because its zeroth-order seminorms are part of the topology. Hence the identity map has closed graph. The source spaces and both target alternatives are Fréchet, so the closed graph theorem supplies continuity automatically. The only technical check is completeness of C^m_0 when $m = \infty$.

Main Theorem

Theorem 1. *Let $K : \Omega\times\Omega \to \mathbb{C}$ be a positive-definite kernel with RKHS $\mathcal{H}_K$, and let $m \in \mathbb{N}_0\cup\{\infty\}$.*

1. *If $\mathcal{H}_K \subset C^m(\Omega)$, then the canonical inclusion*

$$J : \mathcal{H}_K \longrightarrow C^m(\Omega), \qquad Jf = f,$$

   is continuous.

2. *If $\mathcal{H}_K \subset C_0^m(\Omega)$, then the canonical inclusion $J : \mathcal{H}_K \to C_0^m(\Omega)$ is continuous.*

Thus $\mathcal{H}_K \subset C_*^m(\Omega)$ implies $\mathcal{H}_K \hookrightarrow C_*^m(\Omega)$ for both meanings of the star in the source question.

Proof. Write $E = C^m(\Omega)$ in the first case and $E = C_0^m(\Omega)$ in the second. The RKHS $\mathcal{H}_K$ is a Hilbert space, hence a Fréchet space. The space E, with precisely the topology used in the source paper, is also a Fréchet space. For completeness, this standard fact is checked below.

We show that the graph of the linear map $J : \mathcal{H}_K \to E$ is closed. Suppose that a sequence $(f_n) \subset \mathcal{H}_K$ satisfies

$$f_n \longrightarrow f \quad \text{in } \mathcal{H}_K, \qquad Jf_n \longrightarrow g \quad \text{in } E.$$

For every $x \in \Omega$, the reproducing property and Cauchy–Schwarz give

$$|f_n(x) - f(x)| = \left|\langle f_n - f, K(\cdot, x)\rangle_K\right| \le \sqrt{K(x,x)}\, \|f_n - f\|_K \longrightarrow 0.$$

On the other hand, convergence in either target E implies pointwise convergence of the zeroth-order functions, so $f_n(x) \to g(x)$. Therefore $f(x) = g(x)$ for every x, or $Jf = g$. The graph is closed.

The closed graph theorem for Fréchet spaces [3, Chapter 2] now says that J is continuous. □

Why the Target Spaces Are Fréchet

For $C^m(\Omega)$, choose a compact exhaustion $(L_j)_{j\ge1}$ of the open set Ω. Its topology is generated by the countable seminorms

$$p_{j,r}(f) = \max_{|\alpha|\le r} \sup_{x\in L_j} |D^\alpha f(x)|, \qquad j \ge 1, \quad r \le m.$$

For finite m only finitely many derivative orders occur, while for $m = \infty$ the family remains countable. A Cauchy sequence has locally uniform limits for every derivative; the elementary local uniform differentiation theorem shows that these limits are the derivatives of its zeroth-order limit. Hence $C^m(\Omega)$ is complete and metrizable.

For $C_0^m(\Omega)$, use

$$q_r(f) = \max_{|\alpha|\le r} \|D^\alpha f\|_\infty\,, \qquad r \le m.$$

When $m < \infty$, q_m is a complete norm; when $m = \infty$, the countable family (q_r) gives a complete metrizable locally convex topology. Indeed, a Cauchy sequence has uniform limits for all derivatives, the same differentiation argument identifies them correctly, and a uniform limit of functions vanishing at infinity again vanishes at infinity. Thus this target is Fréchet as well.

Quantitative Consequences

The theorem says more concretely that every target seminorm is controlled by the RKHS norm. In the $C^m(\Omega)$ case, for every compact $L \subset \Omega$ and finite $r \le m$, there is a constant $C_{L,r}$ such that

$$\max_{|\alpha|\le r} \sup_{x\in L} |D^\alpha f(x)| \le C_{L,r}\, \|f\|_K \qquad (f \in \mathcal{H}_K).$$

In the $C_0^m(\Omega)$ case, there is for each finite $r \le m$ a global constant C_r such that

$$\max_{|\alpha|\le r} \|D^\alpha f\|_\infty \le C_r \|f\|_K .$$

In particular, every derivative evaluation $f \mapsto D^\alpha f(x)$ is a bounded functional on $\mathcal{H}_K$ and therefore has a Riesz representer in the RKHS.

Audit of the Argument

- RKHS-norm convergence gives pointwise convergence for every kernel, because each point evaluation is bounded by definition. Kernel continuity is not being smuggled into the proof.
- Only zeroth-order pointwise convergence is needed to identify the two limits. The hypothesis that all RKHS elements lie in the target makes the canonical map everywhere defined.
- Both values $m < \infty$ and $m = \infty$ are covered. In the latter case the seminorm families are countable because multi-indices in $\mathbb{N}_0^d$ are countable.
- Vanishing at infinity is stable under uniform limits, so the C_0^m target is complete rather than merely a subspace of a complete product.
- The conclusion is exactly continuity of the canonical inclusion. It does not assert joint (m, m)-smoothness of K, which would require a separate argument.

Literature Check

The run's registry, solution, attempt, and proof-gap indexes were searched for the arXiv id and the phrases "smooth RKHS inclusion", "$\mathcal{H}_K$ subset C^m", and "closed graph". External searches used the exact source sentence and the phrases "RKHS contained in C^m continuously embedded" and "RKHS C^m closed graph theorem". These searches found applications of the same general closed-graph mechanism, but no paper specifically resolving this source question. The novelty verdict is therefore *candidate new direct answer*, subject to specialist review rather than a claim of bibliographic exhaustiveness.

A separate open question in the source paper concerning metrization of weak convergence on Polish spaces is not claimed here: arXiv:2006.09268 explicitly corrected the source theorem and settled that issue negatively outside the compact case [2].

Human-Review Recommendation

Review is recommended for promotion. The decisive checks are narrow: verify that the source's two target topologies are exactly the Fréchet topologies used here, including $m = \infty$, and then verify the two pointwise-limit implications in the closed-graph argument. No computational or conditional dependency remains.

References

[1] C.-J. Simon-Gabriel and B. Schölkopf, *Kernel Distribution Embeddings: Universal Kernels, Characteristic Kernels and Kernel Metrics on Distributions*, J. Mach. Learn. Res. 19 (2018), no. 44, 1–29; arXiv:1604.05251.

[2] C.-J. Simon-Gabriel, A. Barp, B. Schölkopf, and L. Mackey, *Metrizing Weak Convergence with Maximum Mean Discrepancies*, J. Mach. Learn. Res. 24 (2023), no. 184, 1–20; arXiv:2006.09268.

[3] W. Rudin, *Functional Analysis*, second edition, McGraw–Hill, New York, 1991.